Lecture 2

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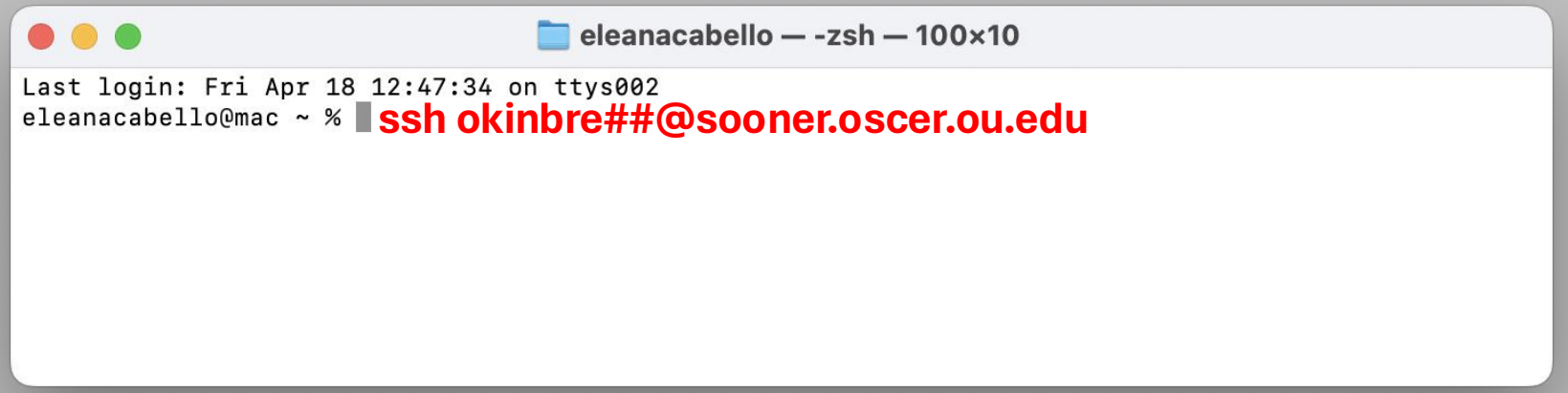
Let’s get on the same page

- Windows systems – Log onto OSCER using MobaXterm
- Mac systems - Find the Terminal application

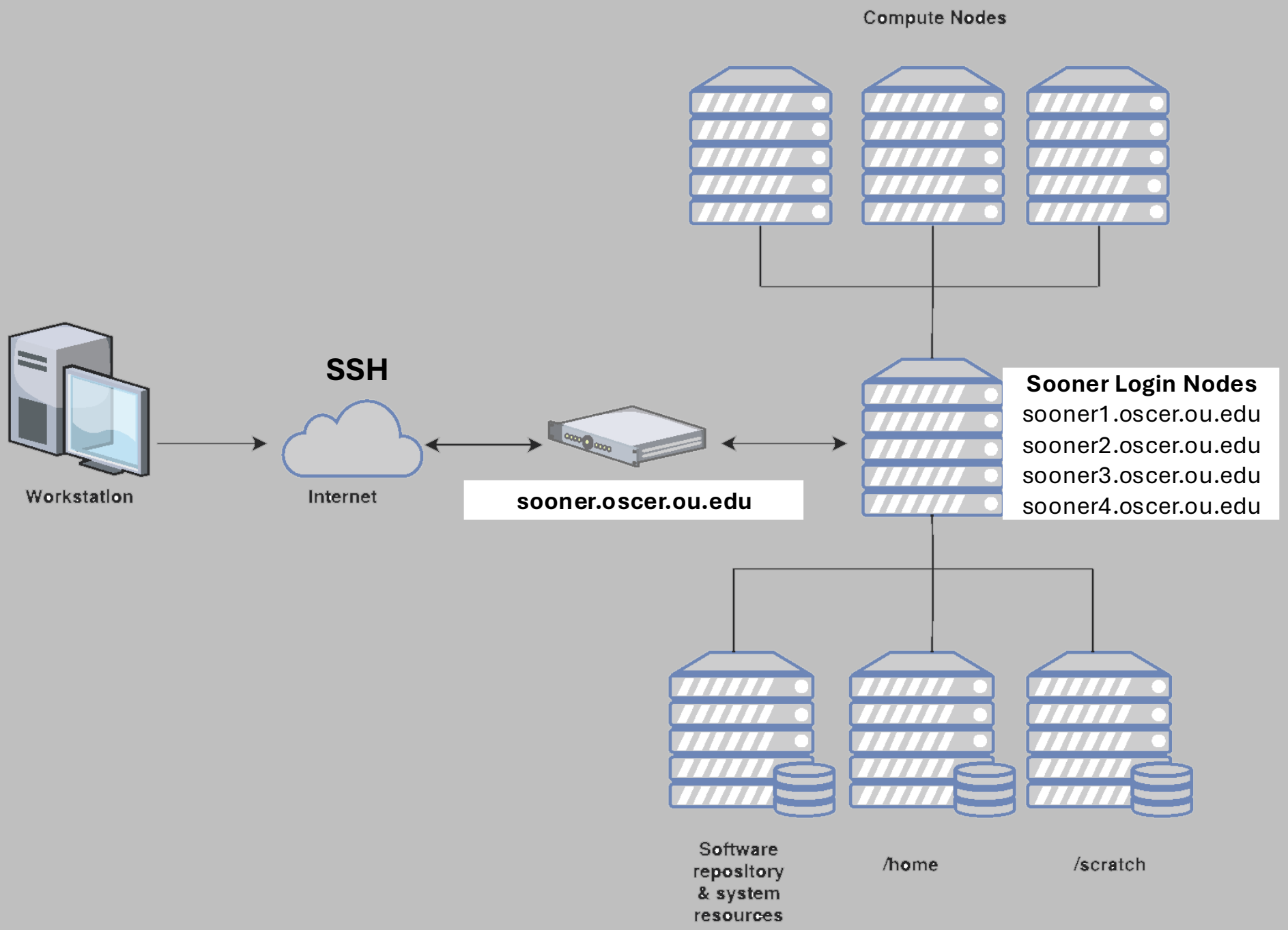
- **Terminal** - A text-based user interface that allows users to interact with a computer by typing commands
- **Directory** - A folder location in your computer
- **Working Directory** - The current directory a user is in
- **Path** - Directions to a folder or file

Let's log into your guest accounts

- Gather you log in information, username and password.



- Type the following command to access you OSCER account: **ssh okinbre##@sooner.oscer.ou.edu**
- It will ask for your password, start typing it. It will not appear on your screen do not be alarmed this is normal. Just make sure to enter it correctly then hit enter.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SSH
Workstation
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Sooner Login Nodes
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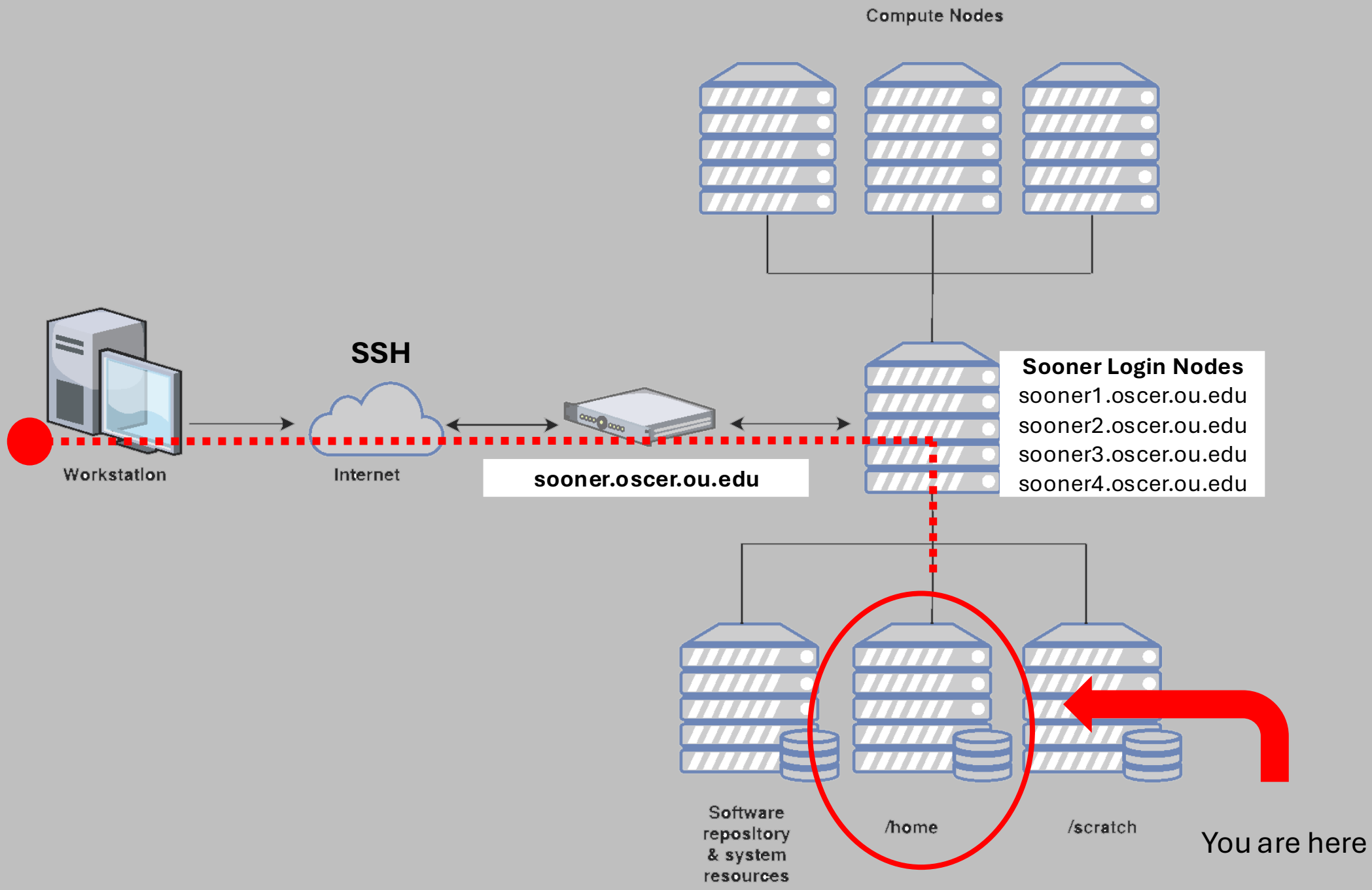
sooner2.oscer.ou.edu
sooner3.oscer.ou.edu
sooner4.oscer.ou.edu
Software repository & system resources
/home
/scratch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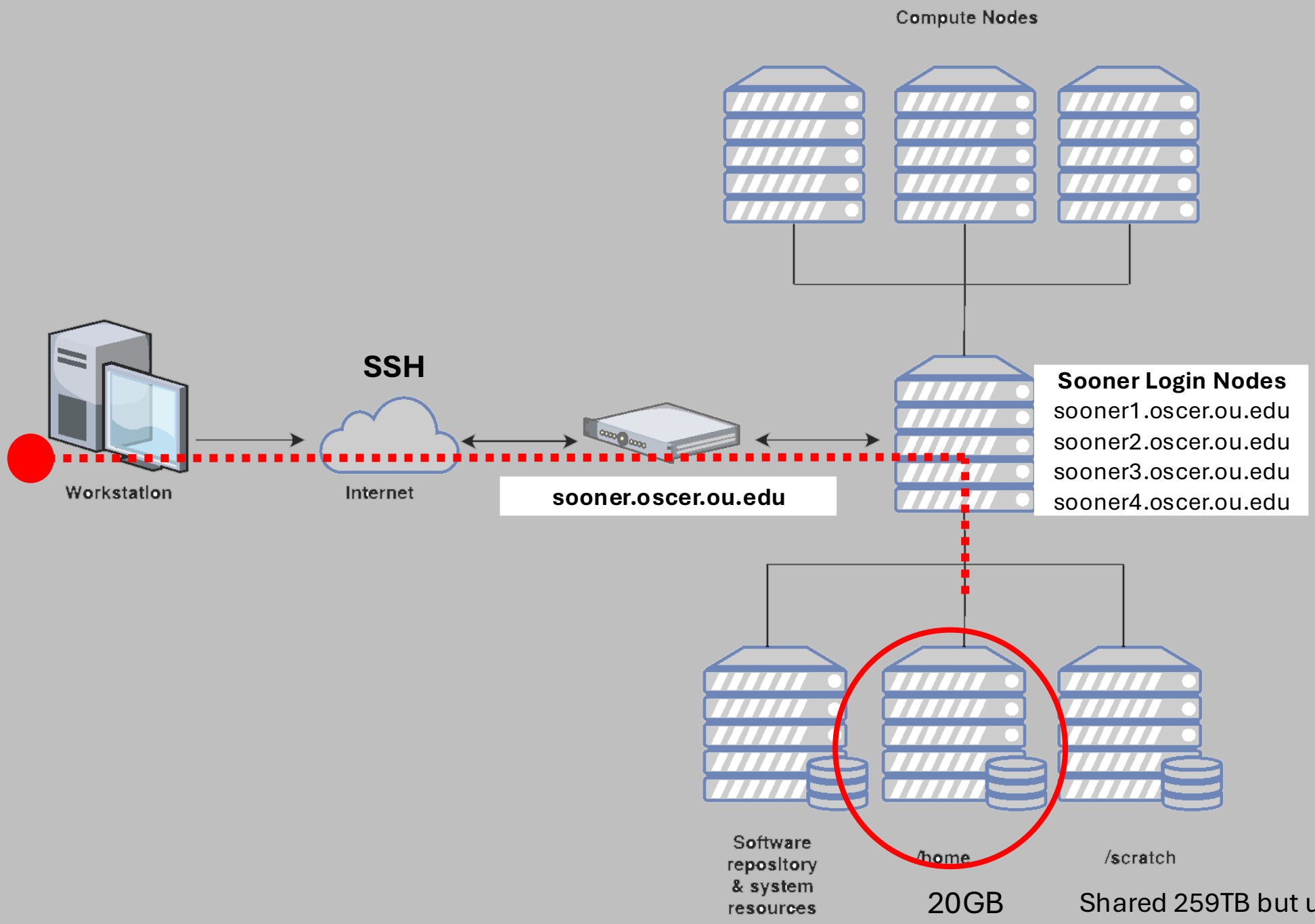
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Software repository & system resources
/home
/scratch
20GB
Shared 259TB but unlimited in a way

BASH Programming

From where you are type: **cp –r /scratch/elec/lecture_2 /scratch/okinbre##**

Now type: **cd /scratch/okinbre##**

Navigating Directories

```
pwd         # Print working directory
ls          # List contents in a directory
ls -lh      # List contents in a directory in a list format and human read able sizes
tree        # List directory and file tree

cd DIR      # Change working directory to the entered directory
cd ..       # Change working directory to parent directory of current directory
cd ~        # Change working directory to home
```

Creating, Moving, and Deleting Directories

```
mkdir DIR      # Create directory
rmdir DIR      # Remove Directory
rm -r DIR      # Remove Directory
mv ONE_DIR TWO_DIR  # Move directory to new directory
```

Deleting, Copying, and Moving Files

```
rm FILE.txt                 # Remove file
cp FILE.txt FILE.txt        # Copy file to new directory
mv FILE.txt FILE.txt        # Move file to new directory
```

There is no back button once things are deleted or overwritten it is gone

Command Line Text Editor

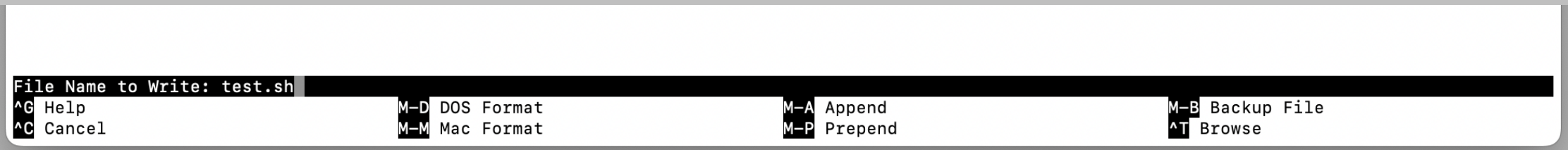
```
nano file_nano_test.txt      # Opens file in line for editing
```

Navigate with up, left, right, and down keys

Type as usual

Exit using CTRL+ X

The above message will appear because you made changes, click Y on your keyboard



It will then ask about the filename you want to save it to, to accept click the ENTER key on your keyboard

```
#!/bin/bash
#SBATCH --partition=normal
#SBATCH --ntasks=1
#SBATCH --mem=1G
#SBATCH --job-name=jobname
#SBATCH --output=jobname_%J_%A_%a_stdout.txt
#SBATCH --error=jobname_%J_%A_%a_stderr.txt
#SBATCH --time=0:10:00
#SBATCH --mail-user=EMAIL@ou.edu
#SBATCH --mail-type=ALL
#SBATCH --chdir=/scratch/USER/lecture_2
```

#SBATCH Directives

Where to send your job
Number of tasks
Memory needed
Name of job
Output file name
Error file name
Time limit
Email to send updates to, **PLEASE REMEMBER TO PUT YOURS**
All notifications
Working directory of the job

```
#==============================================================
# BASH Strict mode (i.e. "fail fast" to reduce hard-to-find bugs)
set -e           # EXIT the script if any command returns non-zero exit status.
set -E           # Make ERR trapping work inside functions too.
set -u           # Variables must be pre-defined before using them.
set -o pipefail # If a pipe fails, returns the error code for the failed pipe
                 #  even if it isn't the last command in a series of pipes.
#==============================================================
```

Quick Exercise

Change into your scratch: **cd /scratch/okinbre##**

Run this: **pwd**

Then type: **nano first_exercise.sh**

Make sure you are in /scratch/user/lecture_2!

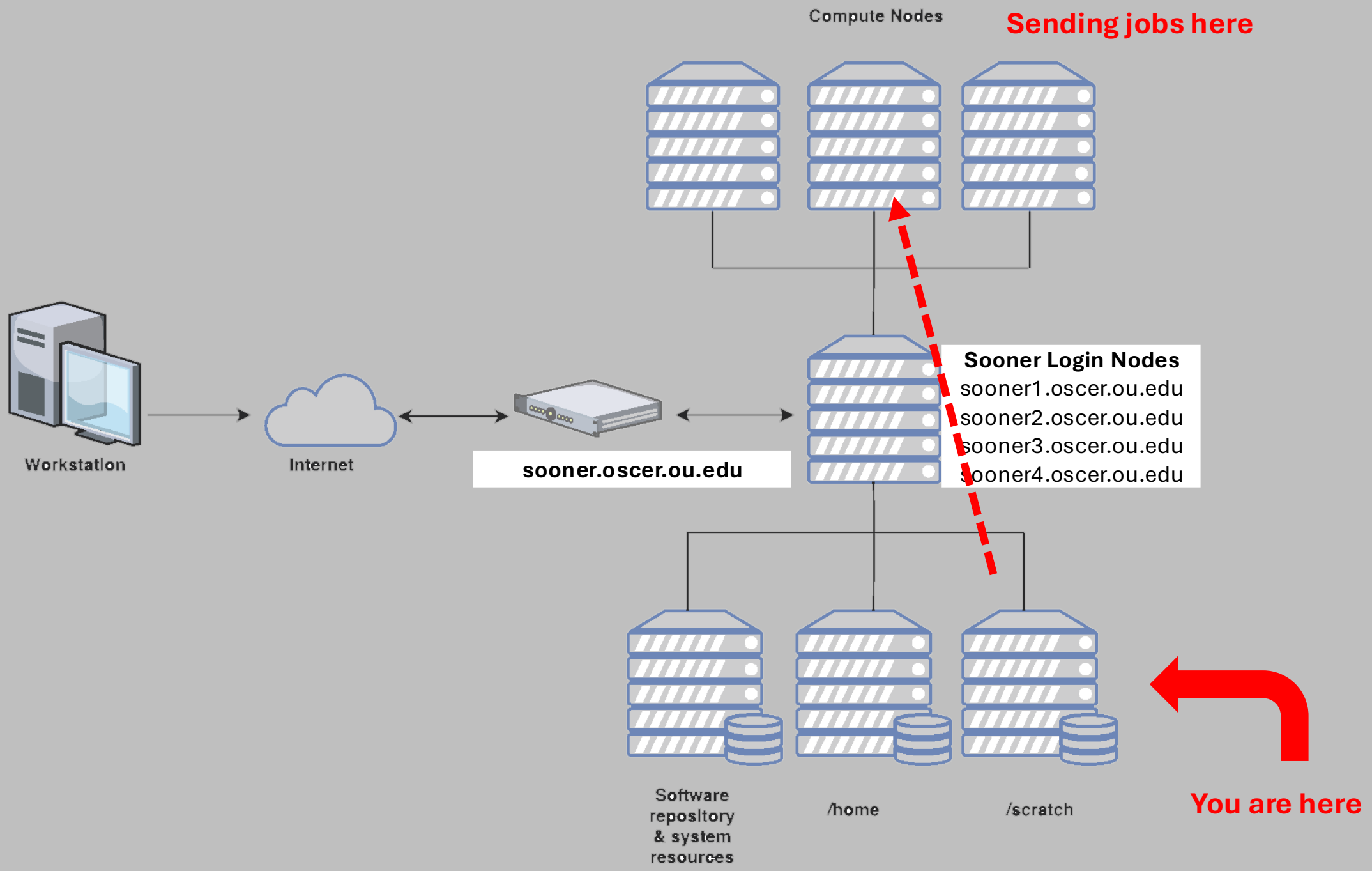
```
#!/bin/bash
#SBATCH --partition=normal
#SBATCH --ntasks=1
#SBATCH --mem=1G
#SBATCH --job-name=jobname
#SBATCH --output=jobname_%J_%A_%a_stdout.txt
#SBATCH --error=jobname_%J_%A_%a_stderr.txt
#SBATCH --time=0:10:00
#SBATCH --mail-user=EMAIL@ou.edu
#SBATCH --mail-type=ALL
#SBATCH --chdir=/scratch/USER/lecture_2

#==============================================================
# BASH Strict mode (i.e. "fail fast" to reduce hard-to-find bugs)
set -e       # EXIT the script if any command returns non-zero exit status.
set -E       # Make ERR trapping work inside functions too.
set -u       # Variables must be pre-defined before using them.
set -o pipefail  # If a pipe fails, returns the error code for the failed pipe
          #  even if it isn't the last command in a series of pipes.
#==============================================================

echo "Hello World!" > helloWorld.txt
```

CHANGE THIS TO YOUR EMAIL as in firstname-lastname@ou.edu

> redirects output into a file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Useful Tips and Questions

Before you leave today

- ❑ Have R and RStudio installed